

ASSIGNMENT:1 Perform the following operations using Python on suitable data sets:
Read data from different formats (like csv, xls) Find Shape of Data Find Missing Values
Find data type of each column Finding out Zero's Indexing and selecting data, sort
data, Describe attributes of data, checking data types of each column, counting
unique values of data, format of each column, converting variable data type (e.g. from
long to short, vice versa)

ROLL NO: 60

DATE: 8-1-2026

NAME: AYUSH PATHAK

STARTED:4:35

ENDED: 5:35

CLASS:CSAID B3

```
from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

```
import numpy as np
import pandas as pd

# path = pd.read_csv("/content/sample_data/california_housing_test.csv")
path = pd.read_json("/content/sample_data/anscombe.json")

print(path)
```

	Series	X	Y
0	I	10	8.04
1	I	8	6.95
2	I	13	7.58
3	I	9	8.81
4	I	11	8.33
5	I	14	9.96
6	I	6	7.24
7	I	4	4.26
8	I	12	10.84
9	I	7	4.81
10	I	5	5.68
11	II	10	9.14
12	II	8	8.14
13	II	13	8.74
14	II	9	8.77
15	II	11	9.26
16	II	14	8.10

```

17    II    6    6.13
18    II    4    3.10
19    II   12    9.13
20    II    7    7.26
21    II    5    4.74
22   III   10    7.46
23   III    8    6.77
24   III   13   12.74
25   III    9    7.11
26   III   11    7.81
27   III   14    8.84
28   III    6    6.08
29   III    4    5.39
30   III   12    8.15
31   III    7    6.42
32   III    5    5.73
33    IV    8    6.58
34    IV    8    5.76
35    IV    8    7.71
36    IV    8    8.84
37    IV    8    8.47
38    IV    8    7.04
39    IV    8    5.25
40    IV   19   12.50
41    IV    8    5.56
42    IV    8    7.91
43    IV    8    6.89

```

```

import numpy as np
import pandas as pd

path = pd.read_csv("/content/sample_data/california_housing_test.csv")

print(path)

```

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	\
0	-122.05	37.37	27.0	3885.0	661.0	
1	-118.30	34.26	43.0	1510.0	310.0	
2	-117.81	33.78	27.0	3589.0	507.0	
3	-118.36	33.82	28.0	67.0	15.0	
4	-119.67	36.33	19.0	1241.0	244.0	
...	...	...	...	...	...	
2995	-119.86	34.42	23.0	1450.0	642.0	
2996	-118.14	34.06	27.0	5257.0	1082.0	
2997	-119.70	36.30	10.0	956.0	201.0	
2998	-117.12	34.10	40.0	96.0	14.0	
2999	-119.63	34.42	42.0	1765.0	263.0	

	population	households	median_income	median_house_value
0	1537.0	606.0	6.6085	344700.0
1	809.0	277.0	3.5990	176500.0
2	1484.0	495.0	5.7934	270500.0
3	49.0	11.0	6.1359	330000.0
4	850.0	237.0	2.9375	81700.0
...	...	...	...	...
2995	1258.0	607.0	1.1790	225000.0

2996	3496.0	1036.0	3.3906	237200.0
2997	693.0	220.0	2.2895	62000.0
2998	46.0	14.0	3.2708	162500.0
2999	753.0	260.0	8.5608	500001.0

[3000 rows x 9 columns]

```
import pandas as pd

s= pd.Series([1,2,3,np.nan,3,"Ayush"])

print(s)
```

```
0      1
1      2
2      3
3     NaN
4      3
5    Ayush
dtype: object
```

```
import pandas as pd

data ={
    "Student_ID": list(range(101, 121)),
    "Name": [
        "Alice", "Bob", "Charlie", "Diana", "Ethan",
        "Fiona", "George", "Hannah", "Ian", "Julia",
        "Kevin", "Laura", "Michael", "Nina", "Oscar",
        "Priya", "Quentin", "Riya", "Sam", "Tina"
    ],
    "Age": [20, 21, 22, 20, 23, 22, np.nan, 23, 20, 22,
            24, 21, 22, np.nan, 20, 22, 21, 23, 24, 22],
    "Gender": [
        "Female", "Male", "Male", "Female", "Male",
        "Female", "Male", "Female", "Male", "Female",
        "Male", "Female", np.nan, "Female", "Male",
        "Female", "Male", "Female", "Male", "Female"
    ],
    "Department": [
        "Computer Science", "Mechanical", "Electrical", "Civil", "Computer S
        "Information Technology", "Mechanical", np.nan, "Electrical", "Compu
        "Mechanical", "Civil", "Information Technology", "Computer Science",
        "Mechanical", "Civil", "Computer Science", "Information Technology",
    ],
    "Marks": [85, 78, 92, 74, 88, 81, 76, 90, np.nan, 87,
              69, 95, 72, 89, 91, 77, 84, 93, 68, np.nan],
    "Attendance": [
        "90%", "85%", "95%", "80%", "88%", "92%", "84%", "89%", "91%", "87%"
        "75%", "96%", "82%", "93%", np.nan, "79%", "88%", "97%", "76%", "90%"
    ],
    "Internship": [
        "Yes", "No", "Yes", "No", "Yes", "No", "Yes", "No", "Yes", "No",
        "Yes", "No", "Yes", "No", "Yes", np.nan, "Yes", "No", "Yes", "No"
```

```
]
}
```

```
df= pd.DataFrame(data)
df
```

	Student_ID	Name	Age	Gender	Department	Marks	Attendance	Internsh:
0	101	Alice	20.0	Female	Computer Science	85.0	90%	Y
1	102	Bob	21.0	Male	Mechanical	78.0	85%	M
2	103	Charlie	22.0	Male	Electrical	92.0	95%	Y
3	104	Diana	20.0	Female	Civil	74.0	80%	M
4	105	Ethan	23.0	Male	Computer Science	88.0	88%	Y
5	106	Fiona	22.0	Female	Information Technology	81.0	92%	M
6	107	George	NaN	Male	Mechanical	76.0	84%	Y
7	108	Hannah	23.0	Female	NaN	90.0	89%	M
8	109	Ian	20.0	Male	Electrical	NaN	91%	Y
9	110	Julia	22.0	Female	Computer Science	87.0	87%	M
10	111	Kevin	24.0	Male	Mechanical	69.0	75%	Y
11	112	Laura	21.0	Female	Civil	95.0	96%	M
12	113	Michael	22.0	NaN	Information Technology	72.0	82%	Y
13	114	Nina	NaN	Female	Computer Science	89.0	93%	M
14	115	Oscar	20.0	Male	Electrical	91.0	NaN	Y
15	116	Priya	22.0	Female	Mechanical	77.0	79%	Na
16	117	Quentin	21.0	Male	Civil	84.0	88%	Y

Next steps:

[Generate code with df](#)[New interactive sheet](#)

df

	Student_ID	Name	Age	Gender	Department	Marks	Attendance	Internsh:
0	101	Alice	20.0	Female	Computer Science	85.0	90%	Y
1	102	Bob	21.0	Male	Mechanical	78.0	85%	M
2	103	Charlie	22.0	Male	Electrical	92.0	95%	Y
3	104	Diana	20.0	Female	Civil	74.0	80%	M
4	105	Ethan	23.0	Male	Computer Science	88.0	88%	Y
5	106	Fiona	22.0	Female	Information Technology	81.0	92%	M
6	107	George	NaN	Male	Mechanical	76.0	84%	Y
7	108	Hannah	23.0	Female	NaN	90.0	89%	M
8	109	Ian	20.0	Male	Electrical	NaN	91%	Y
9	110	Julia	22.0	Female	Computer Science	87.0	87%	M
10	111	Kevin	24.0	Male	Mechanical	69.0	75%	Y
11	112	Laura	21.0	Female	Civil	95.0	96%	M
12	113	Michael	22.0	NaN	Information Technology	72.0	82%	Y
13	114	Nina	NaN	Female	Computer Science	89.0	93%	M
14	115	Oscar	20.0	Male	Electrical	91.0	NaN	Y
15	116	Priya	22.0	Female	Mechanical	77.0	79%	Na
16	117	Quentin	21.0	Male	Civil	84.0	88%	Y

```
import numpy as np
import pandas as pd

data=pd.read_csv("/content/orders.csv")

print("\n\nThe confugiration of dataset is: ")
data.shape
```

```
The confugiration of dataset is:
(1000, 7)
```

```
import numpy as np
import pandas as pd

data=pd.read_csv("/content/orders.csv")
```

```
print("\n\nThe missing values of the dataset are:\n\n ")
data.isnull()
```

The missing values of the dataset are:

	order_id	order_date	customer_id	product_id	qty	unit_price	currency
0	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...
995	False	False	False	False	False	False	False
996	False	False	False	False	False	False	False
997	False	False	False	False	False	False	False
998	False	False	False	False	False	False	False
999	False	False	False	False	False	False	False

1000 rows × 7 columns

```
import numpy as np
import pandas as pd

data=pd.read_csv("/content/orders.csv")

print("\n\nThe total missing values of the dataset are\n\n : ")
data.isnull().sum()
```

The total missing values of the dataset are

:

	0
order_id	0
order_date	0
customer_id	0
product_id	0
qty	0
unit_price	0
currency	0

dtype: int64

```
import numpy as np
import pandas as pd
```

```
data=pd.read_csv("/content/orders.csv")
```

```
df= pd.to_numeric(data['unit_price'],errors='coerce')
print("\n\nAll the datatypes present are:\n\n ")
data.dtypes
```

All the datatypes present are:

	0
order_id	int64
order_date	object
customer_id	int64
product_id	int64
qty	int64
unit_price	float64
currency	object

dtype: object

```
import numpy as np
import pandas as pd

data=pd.read_csv("/content/orders.csv")

print("\n\n Finding values with price above 200000\n\n")
(df>=200000)
```

Finding values with price above 200000

	unit_price
0	True
1	True
2	True
3	True
4	True
...	...
995	True
996	True
997	False
998	True
999	False

1000 rows × 1 columns

dtype: bool

```
import numpy as np
import pandas as pd

data=pd.read_csv("/content/orders.csv")

print("\n\n Finding total values : \n\n")
(df>=200000).sum()
```

Finding total values :

np.int64(807)


```
data.describe()
```

	order_id	customer_id	product_id	qty	unit_price
count	1000.000000	1000.000000	1000.000000	1000.000000	1.000000e+03
mean	500.500000	538.300000	91.044000	1.288000	1.959816e+07
std	288.819436	256.609006	52.415527	0.547134	2.482369e+07
min	1.000000	100.000000	1.000000	1.000000	8.000000e+00
25%	250.750000	320.750000	45.000000	1.000000	3.775000e+06
50%	500.500000	534.000000	91.500000	1.000000	1.123000e+07
75%	750.250000	750.250000	137.000000	1.000000	2.590750e+07
max	1000.000000	998.000000	180.000000	3.000000	2.503100e+08

```
data.sort_values(by='customer_id')
```

	order_id	order_date	customer_id	product_id	qty	unit_price	currency
552	553	2025-11-02	100	93	1	21230000.0	IRR
596	597	2025-11-06	101	30	3	1550000.0	IRR
177	178	2025-10-05	103	151	1	6960000.0	IRR
150	151	2025-10-04	104	38	1	35930000.0	IRR
753	754	2025-11-17	106	138	1	4020000.0	IRR
...	...	...	...	...	...	...	...
187	188	2025-10-06	992	87	2	733.5	USD
7	8	2025-09-23	992	123	1	8440000.0	IRR
786	787	2025-11-20	993	75	1	55210000.0	IRR
704	705	2025-11-13	997	71	1	8520000.0	IRR
312	313	2025-10-15	998	48	1	30660000.0	IRR

1000 rows × 7 columns

Double-click (or enter) to edit

```
df.sort_values(by='Age', ascending=True)
```

	Student_ID	Name	Age	Gender	Department	Marks	Attendance	Internsh:
0	101	Alice	20.0	Female	Computer Science	85.0	90%	Y
3	104	Diana	20.0	Female	Civil	74.0	80%	M
8	109	Ian	20.0	Male	Electrical	NaN	91%	Y
14	115	Oscar	20.0	Male	Electrical	91.0	NaN	Y
11	112	Laura	21.0	Female	Civil	95.0	96%	M
16	117	Quentin	21.0	Male	Civil	84.0	88%	Y
1	102	Bob	21.0	Male	Mechanical	78.0	85%	M
2	103	Charlie	22.0	Male	Electrical	92.0	95%	Y
15	116	Priya	22.0	Female	Mechanical	77.0	79%	Na
12	113	Michael	22.0	NaN	Information Technology	72.0	82%	Y
5	106	Fiona	22.0	Female	Information Technology	81.0	92%	M
9	110	Julia	22.0	Female	Computer Science	87.0	87%	M
19	120	Tina	22.0	Female	Electrical	NaN	90%	M
7	108	Hannah	23.0	Female	NaN	90.0	89%	M
17	118	Riya	23.0	Female	Computer Science	93.0	97%	M
4	105	Ethan	23.0	Male	Computer Science	88.0	88%	Y
10	111	Kevin	24.0	Male	Mechanical	69.0	75%	Y

Taking info about the data frame(data types)

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Student_ID      20 non-null    int64
1   Name            20 non-null    object
2   Age             18 non-null    float64
3   Gender          19 non-null    object
4   Department      19 non-null    object
5   Marks           18 non-null    float64
6   Attendance      19 non-null    object
7   Internship      19 non-null    object
dtypes: float64(2), int64(1), object(5)
```

memory usage: 1.4+ KB

Converting data types

```
df['Marks']=df['Marks'].astype(float)
```

unique values count

```
df.nunique()
```

	0
Student_ID	20
Name	20
Age	5
Gender	2
Department	5
Marks	18
Attendance	17
Internship	2

dtype: int64

HEAD of DF

```
df.head()
```

	Student_ID	Name	Age	Gender	Department	Marks	Attendance	Internship
0	101	Alice	20.0	Female	Computer Science	85.0	90%	Yes
1	102	Bob	21.0	Male	Mechanical	78.0	85%	No
2	103	Charlie	22.0	Male	Electrical	92.0	95%	Yes
3	104	Diana	20.0	Female	Civil	74.0	80%	No

Next steps:

[Generate code with df](#)[New interactive sheet](#)

TAIL OF DF

```
df.tail()
```

	Student_ID	Name	Age	Gender	Department	Marks	Attendance	Internsh:
15	116	Priya	22.0	Female	Mechanical	77.0	79%	Na
16	117	Quentin	21.0	Male	Civil	84.0	88%	Ye
17	118	Riya	23.0	Female	Computer Science	93.0	97%	M
18	119	Sam	24.0	Male	Information Technology	68.0	76%	Ye

TRANSPOSING THE DATA

```
# This is formatted as code
```

df.T

	0	1	2	3	4	5	
Student_ID	101	102	103	104	105	106	
Name	Alice	Bob	Charlie	Diana	Ethan	Fiona	Ge
Age	20.0	21.0	22.0	20.0	23.0	22.0	
Gender	Female	Male	Male	Female	Male	Female	
Department	Computer Science	Mechanical	Electrical	Civil	Computer Science	Information Technology	Mecha
Marks	85.0	78.0	92.0	74.0	88.0	81.0	
Attendance	90%	85%	95%	80%	88%	92%	
Internship	Yes	No	Yes	No	Yes	No	

SELECTING COLUMNS

```
df.loc[:, ["Age", "Marks"]]
```

	Age	Marks
0	20.0	85.0
1	21.0	78.0
2	22.0	92.0
3	20.0	74.0
4	23.0	88.0
5	22.0	81.0
6	NaN	76.0
7	23.0	90.0
8	20.0	NaN
9	22.0	87.0
10	24.0	69.0
11	21.0	95.0
12	22.0	72.0
13	NaN	89.0
14	20.0	91.0
15	22.0	77.0
16	21.0	84.0
17	23.0	93.0
18	24.0	68.0
19	22.0	NaN



DROPPING MISSING VALUES

```
df.dropna()
```

	Student_ID	Name	Age	Gender	Department	Marks	Attendance	Internsh:
0	101	Alice	20.0	Female	Computer Science	85.0	90%	Ye
1	102	Bob	21.0	Male	Mechanical	78.0	85%	M
2	103	Charlie	22.0	Male	Electrical	92.0	95%	Ye
3	104	Diana	20.0	Female	Civil	74.0	80%	M
4	105	Ethan	23.0	Male	Computer Science	88.0	88%	Ye
5	106	Fiona	22.0	Female	Information Technology	81.0	92%	M
9	110	Julia	22.0	Female	Computer Science	87.0	87%	M
10	111	Kevin	24.0	Male	Mechanical	69.0	75%	Ye
11	112	Laura	21.0	Female	Civil	95.0	96%	M
16	117	Quentin	21.0	Male	Civil	84.0	88%	Ye

Getting the nan values

```
df.isna()
```

	Student_ID	Name	Age	Gender	Department	Marks	Attendance	Internship
0	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False
5	False	False	False	False	False	False	False	False

Merging data frames.

```
left = pd.DataFrame({"key": ["foo", "foo"], "lval": [1, 2]})
right = pd.DataFrame({"key": ["foo", "foo"], "rval": [4, 5]})
pd.merge(left, right, on="key")
```

0	False	False	False	False	False	False	False	False
10	key	lval	rval		False	False	False	False
0	foo	1	4	se	False	False	False	False
12	foo	1	5	False	True	False	False	False